

DEEPAK PATKAR

PhD, Theoretical and Computational Chemistry

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Experience	Post-doctoral Fellow	2025-
	Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences	
	Research Topic: <i>CuPiLLS- Copper-Peptide mediated Liquid-Liquid Phase Separation.</i>	
	Senior Associate at Innodata India Private Limited	2024-2025
	Role: Subject Matter Expert Responsibilities: Image-based LLM training data curation	
Academics	Doctor of Philosophy, Chemistry	2019-2024
	Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India	
	Thesis Supervisor: Dr. Milind Madhusudan Deshmukh [link]	
	Thesis: <i>Quantum Chemical Studies on the Mixed Molecular Clusters: Quantitative Estimates of Hydrogen Bond Strength and Cooperativity.</i> [link]	
	Master of Science, Chemistry	2017-2019
	Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India	
	Project Supervisor: Dr. Abhilasha Durgavanshi [link]	
	Hands-on experience in HPLC and Gas Chromatography.	
	Bachelor of Science, Triple major in Physics, Chemistry, Criminology and Forensic Science	2014-2017
	Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India	
	Built foundational understanding of principles, techniques across diverse disciplines.	
Research Interest	<i>Non-covalent intermolecular interactions and their cooperative effects, transition metal catalysis, machine learning with applications in cheminformatics and bioinformatics.</i>	
Publications	Total Publication = 8 , h-index = 5 , i10-index = 5 , Citations = 131 cf. Google Scholar Id DxGUFX0AAAAJ	
	1. Ismail, T. M.; Patkar, D.; Sajith, P. K.; Deshmukh, M. M. Hydrogen bond strengths in microhydrated clusters of HNO and HONO: Energetic insights via molecular tailoring approach <i>New J. Chem.</i> 2024, 48, 11591–11605. [link] 2. Ismail, T. M.; Patkar, D.; Sajith, P. K.; Deshmukh, M. M. Interplay of Hydrogen, Pnicogen and Chalcogen Bonding in X(H₂O)_n=1-5 (X = NO, NO⁺, and NO⁻) Complexes: Energetics Insights via Molecular Tailoring Approach <i>J. Phys. Chem. A</i> 2023, 127,10360–10374. [link] 3. Patkar, D.; Chopra, D.; Deshmukh, M. M. Characterization of non-covalent contacts in mono- and di-halo substituted acetaldehydes: probing the substitution effects of electron donating and withdrawing groups <i>Phys. Chem. Chem. Phys.</i> 2023, 25, 2946. [link] 4. Patkar, D.; Ahirwar, M. B.; Deshmukh, M. M. A Tug of War between the Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. <i>ChemPhysChem</i> 2022, 23, e202200476. [link] 5. Patkar, D.; Ahirwar, M. B.; Deshmukh, M. M. Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach <i>ChemPhysChem</i> 2022, 23, e202200143. [link] 6. Patkar, D.; Ahirwar, M. B.; Shrivastava, S. P.; Deshmukh, M. M. Assessment of Hydrogen Bond Strengths and Cooperativity in Self- and Cross-Associating Cyclic (HF)_m(H₂O)_n (m + n = 2 to 8) Clusters. <i>New J. Chem.</i> 2022, 46, 2368–2379. [link] 7. Patkar, D.; Ahirwar, M. B.; Gadre, S. R.; Deshmukh, M. M. Unusually Large Hydrogen-Bond Cooperativity in Hydrogen Fluoride Clusters, (HF)_n, n = 3 to 8, Revealed by the Molecular Tailoring Approach. <i>J. Phys. Chem. A</i> 2021, 125, 8836–8845. [link] 8. Ahirwar, M. B.; Patkar, D.; Yadav, I.; Deshmukh, M. M. Appraisal of Individual Hydrogen Bond Strengths and Cooperativity in Ammonia Clusters via Molecular Tailoring Approach. <i>Phys. Chem. Chem. Phys.</i> 2021, 23, 17224–17231. [link] 9. Patkar, D. Uncovering the Individual Hydrogen Bond Strengths and Cooperativity in Pure (H₂S)_n and Mixed (H₂O)_m(H₂S)_n (m + n = 2 to 8) Clusters. <i>ChemRxiv</i> 2025. [link] [Pre-print] 10. Patkar, D.; Deshmukh, M. M.; Chopra, D. Quantitative Examination of C-H...F contacts in 2, 2, 2-Trifluoroacetophenone: Unravelling the Substitution Effects on Interaction Strength. <i>Chem. Comm.</i> 2026. (to be submitted soon)	

Selected Talks and Presentations

1. National Seminar on "Recent Advances in Chemical Sciences", Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March, **2023**. [\[link\]](#)
Title: Competition between Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. [\[link\]](#)
2. DAE Symposium on "Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, **2022**. [\[link\]](#)
Title: Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach. [\[link\]](#)
3. DAE Symposium on "Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, **2021**. [\[link\]](#)
Title: Evaluation of individual Hydrogen Bond Strengths and Cooperativity in $(\text{HF})_m(\text{H}_2\text{O})_n$, $(m + n) = 2$ to 8 Clusters. [\[link\]](#)
4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, December, **2021**. [\[link\]](#)
Title: Estimation of Individual Hydrogen Bond Strengths and Cooperativity in $(\text{HF})_m(\text{H}_2\text{O})_n$, $(m + n) = 2$ to 8 Clusters via Molecular Tailoring Approach. [\[link\]](#)
5. International Seminar, Dr. Harisingh Gour Vishwavidyalaya Sagar, January, **2021**.
Title: Theoretical Study of Hydrosilylation of Olefin Catalyzed by Cationic Calcium Hydride Catalyst. [\[link\]](#)

Selected Conferences and Workshops

1. National Seminar on "Recent Advances in Chemical Sciences", Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March **2023**. [\[link\]](#)
2. DAE Symposium on "Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September **2022**. [\[link\]](#)
3. Hand on Workshop "Molecular Tailoring Approach", Department of Scientific Computing, Modelling and Simulation, Savitribai Phule Pune University, May **2022**. [\[link\]](#)
4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, Kalyani University, December, **2021**. [\[link\]](#)
5. DAE Symposium on "Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September **2021**. [\[link\]](#)

Skills

1. Molecule modelling software: ChemCraft, Chemdraw, Gaussview, Avogadro, Spartan.
2. Ab initio QM Software: Gaussian, ORCA, GAMESS, Psi4, Material Studio, DAMQT, AIMALL, Multiwfn.
3. Visualization & Analysis Tools: VMD, PyMOL, ChimeraX, VESTA, OVITO.
4. Molecular Dynamics Simulation Software: GROMACS, AMBER TOOL25, CHARMM, LAMMPS.
5. Solid-State Software: VASP, Quantum ESPRESSO, CP2K, Material Studio.
6. Applications Software: LaTeX, GNUplot, Excel, Origin Pro.
7. Programming languages: Fortran, Bash, Python.
8. Other Tools: RDkit, PySCF, machine learning including chemo- and bio- informatics.

Academic Achievements

1. Received INSPIRE scholarship, Department of Science and Technology, India, **2014**.
2. Best Oral Presentation, Departmental Seminar, DHSGSU, India, **2023**.
3. Membership of American Chemical Society, **2021**.

Teaching Experience

- Teaching Assistant at Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP, India
1. Designed and conducted practical chemistry classes for UG students.
 2. Developed computational chemistry lab assignments and guided lab sessions.

References

1. Dr. Milind M. Deshmukh (Ph.D. Supervisor)
Associate Professor, Department of Chemistry, University of Delhi, Delhi
Mobile: +91 8989180813, Email: milind.deshmukh@gmail.com webpage: [\[link\]](#)
2. Prof. Deepak Chopra (Collaborator)
Crystallography and Crystal Chemistry Laboratory, Department of Chemistry, IISER Bhopal
Office: +91-755-669-1311; Fax: +91 755 6692392, Email: dchopra@iiserb.ac.in webpage: [\[link\]](#)
3. Dr. Pookkottu K. Sajith (Collaborator)
Assistant Professor, Department of Chemistry, Farook College, Kozhikode, Kerala
Mobile: +91 9349280803, Email: psajith@farookcollege.ac.in webpage: [\[link\]](#)